

Training Deep Network Models for Fingerprint Image Classification

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Abstract Our investigation aims to answer the research question is it possible to train deep network models that can be re-used to classify a new coming dataset of fingerprint images without re-training the new deep network model?. For this purpose, we collect real datasets of fingerprint images from students at the Can Tho University. After that, we propose to train recent deep networks, such as VGG, ResNet50, Inception-v3, Xception, on the the training dataset with 9,236 fingerprint images of 441 students, to create deep network models. And then, we re-use these resulting deep network models as the feature extraction and only fine-tune the last layer in deep network models for the new fingerprint image datasets. The empirical test results on three real fingerprint image datasets (FP-235, FP-389, FP-559) show that deep network models achieve at least the accuracy of 96.72% on the testsets. Typically, the ResNet50 models give classification accuracy of 99.00%, 98.33%, 98.05% on FP-235, FP-389 and FP-559, respectively.

Keywords: Fingerprint image classification · Deep network · Transfer learning.

1 Introduction

The recognition of fingerprint images is one of the most popular and useful method for identifying individuals due to its uniqueness and durability over time. The recognition of fingerprint images is successfully applied to the applications, like suspect and victim identifications, the recovery of partial fingerprints from a crime scene in forensic science, border control, employment background checks, and secure facility entrance [17,18,26].

In this paper, we are interested in the research problem, in which is it possible to train deep network models for classifying a new coming dataset of fingerprint images without retraining the new deep network from scratch? For this aim, we study to train deep network models for the classification of any new coming dataset of fingerprint images. We start with the collection of real fingerprint image datasets from the students at the Can Tho University. After that, we propose

to train recent deep networks, including VGG [35], ResNet50 [15], Inception-v3 [37], Xception [7], on the training dataset with 9,236 fingerprint images of 441 students, to create deep network models. And then, these resulting deep network models can be re-used as the feature extraction and being only fine-tuned by training the last layer in deep network models for the new fingerprint image datasets. The empirical test results on three real fingerprint image datasets (FP-235, FP-389, FP-559 [9]) show that deep network models achieve at least the accuracy of 96.72% on the testsets without re-training deep network models from scratch. An example of effectiveness obtained by ResNet50 models is illustrated in the classification accuracy of 99.00%, 98.33%, 98.05% on FP-235, FP-389 and FP-559, respectively.

The remainder of this paper is organized as follows. Section 2 illustrates our proposal to train deep network models for fingerprint image classification. Section 3 shows the experimental results before conclusions and future work presented in section 4.

2 Training deep network models for classifying fingerprint images

Building a system for fingerprint image classification bases on the usual framework for the classification of images. It consists of three main works as follows:

1. collecting the dataset of fingerprint images,
2. extracting visual features from fingerprint images and representing them,
3. training classifiers.

As illustrated in [17,18,26], the fingerprint recognition approaches are classified into two categories. First classical approaches commonly use minutiae (e.g. ridge ending, ridge bifurcation, etc.) and the matching method between minutiae of fingerprint images [5,17,18,26,27]. Some classical approaches [13,10,21] also use handcrafted features, like the scale-invariant feature transform (SIFT [23,24]) and the bag-of-words model (BoW [2,20,36]) and training support vector machines (SVM [38]).

Recent approaches [34,3,31,30,28,29,11,4] aim to train deep convolutional neural networks (CNN [19]) or fine-tuning pre-trained deep neural networks (VGG16, VGG19 [35], ResNet50 [15], Inception-v3 [37], Xception [6]) to classify fingerprint images. These CNN-based approaches illustrated in Figure 1 benefits from the ability to learn visual features (low-level, mid-level, high-level) from images and the softmax classifier in an unified framework.

We are interested in training deep network models to classify a new coming dataset of fingerprint images without retraining the new deep network from scratch.

2.1 Dataset collection

We start with the collection of fingerprint image datasets. By supporting from students and colleagues at the College of Information Technology, Can Tho Uni-

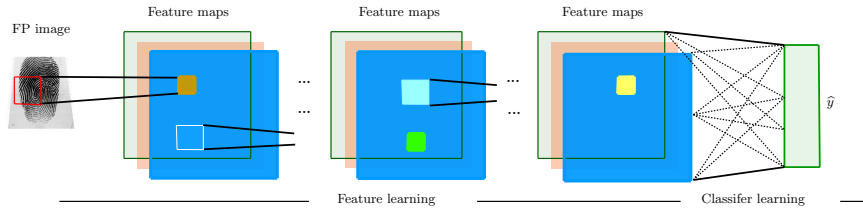


Figure 1. Convolutional neural network (CNN) for classifying fingerprint images

versity, we captured a dataset (named **trainbase**), having 9,236 fingerprint images of 441 individuals with Microsoft Fingerprint Reader (optical fingerprint scanner, resolution: 512 DPI, image size: 355x390, colors: 256 levels grayscale). This fingerprint image dataset is used to train deep network models. And then, we use three real fingerprint datasets named FP-235, FP-389, FP-559 (collected in our previous works [12,11,9], including fingerprint images of 235, 389, and 559 individuals respectively) to evaluate resulting deep network models.

2.2 Deep networks

In last years, the researchers in computer vision and machine learning communities proposed deep neural networks including VGG [35], ResNet50 [15], Inception-v3 [37], Xception [7] that achieve the most accuracy for the classification of ImageNet dataset [8]. We also propose to use these deep network architectures to train models for fingerprint images.

VGG16, VGG19: The VGG network architecture proposed by [35] includes two parts. The first part stacks several VGG block modules (convolutional and pooling layers as shown in Figure 2) to learn visual features from images. The second part consists of fully-connected layers to train classifiers. The VGG architecture tries to capture the spatial information from images. VGG-16, VGG-19 consist of 16 and 19 VGG block modules for the large scale image classification of ImageNet dataset.

Inception-v3: The Inception-v3 network proposed by [37] stacks 11 inception modules (in Figure 3) and global average pooling to learn multi-level, invariant features and the spatial information from images for the classification.

ResNet50: The well-known failure of deep networks is the vanishing gradient problem during the learning process of deep network. To overcome this issue, He and his colleagues [15] proposed the ResNet50 network architecture, using the residual block (in Figure 4) to develop extremely deep networks for classifying ImageNet dataset.

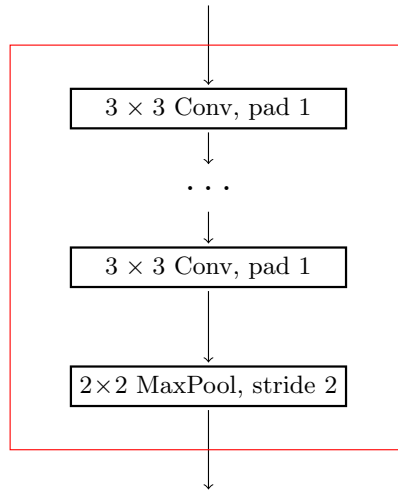


Figure 2. VGG block module

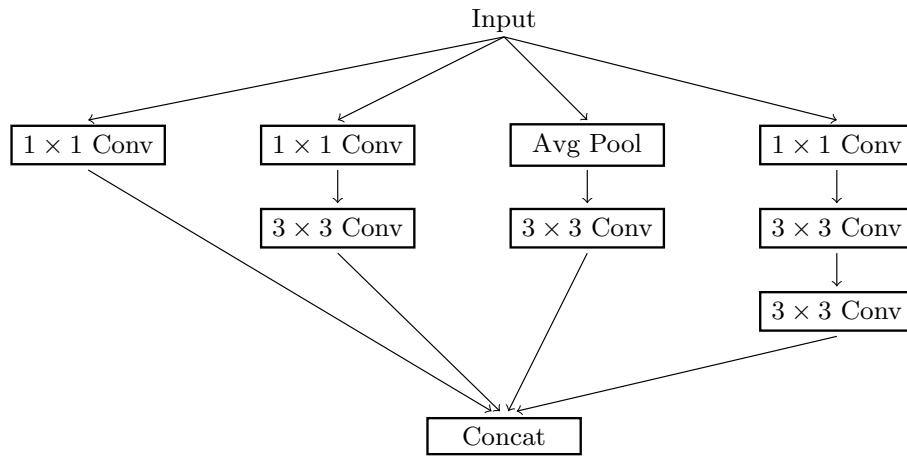


Figure 3. Inception module

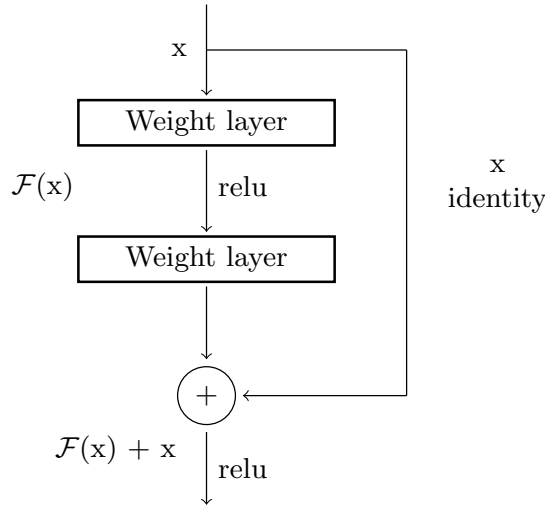


Figure 4. Residual block

Xception: The Xception network proposed by [6] is an extension of the Inception architecture. The Xception module (Figure 5) replaces the standard depthwise separable convolution (the depthwise convolution followed by a pointwise convolution) in Inception modules with depthwise separable convolutions, making the classification improvement.

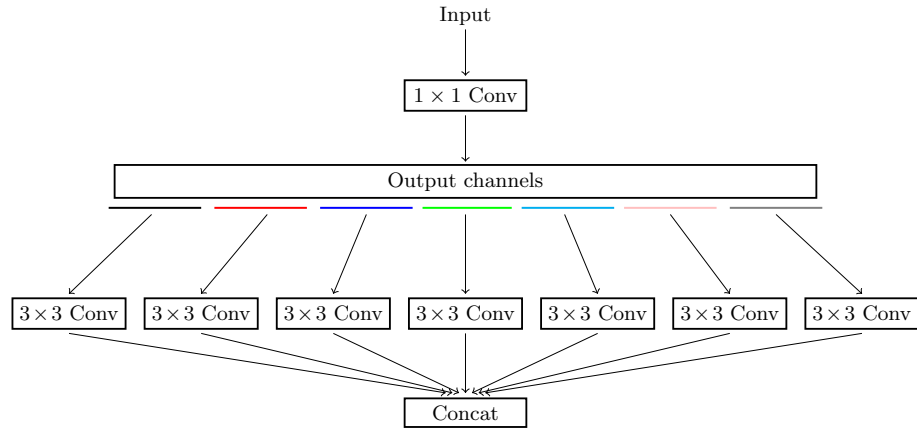
2.3 Training deep network models for classifying fingerprint images

We propose to train deep neural networks, VGG , ResNet50, Inception-v3, Xception on **trainbase** dataset with 9,236 fingerprint images of 441 individuals. We obtain deep network models for this fingerprint image dataset.

Although the deep network models give most accurate classification results but the training task requires big-data, vast computational cost and time resources to learn a new model for the new coming fingerprint image dataset.

In last years, the deep learning community pays attention to the ability to re-use pre-trained models from the source learner in the target task. This strategy is well-known as transfer learning [14]. The main idea is to re-use a pre-trained model on a problem similar to the target problem as the starting point for learning a new model on the target problem [33,39].

Our main idea is to re-use these deep network models being pre-trained on **trainbase** dataset as feature extractors from images. The training task for a new coming fingerprint image dataset is only to update the weights of the last layer while keeping unchanged the first other layers in networks. Therefore, it reduces the training resources and complexity for handling a new coming dataset due to the leverage of pre-trained VGG, ResNet50, Inception-v3, Xception on **trainbase** dataset.

**Figure 5.** Xception module

3 Experimental results

In this section, we present experimental results of deep network models for classifying fingerprint images. We implement the training program in Python using library Keras [6] with backend Tensorflow [1], library Scikit-learn [32] and library OpenCV [16].

All experiments are conducted on a machine Linux Fedora 34, Intel(R) Core i7-4790 CPU, 3.6 GHz, 4 cores and 16 GB main memory and the Gigabyte GeForce RTX 2080Ti 11GB GDDR6, 4352 CUDA cores.

As described in section 2, the starting models are pre-trained VGG, ResNet50, Inception-v3, Xception on **trainbase** dataset with learning rate = 0.001 and number of epochs = 200. The training task for a new fingerprint image dataset is only to fine-tune the last layer in these resulting VGG, ResNet50, Inception-v3, Xception with learning rate = 0.001 and number of epochs = 100.

We propose to use three real fingerprint image datasets (in table 1) to evaluate the classification performance of deep networks. Datasets are randomly split into the trainset (80% fingerprint images) and the testset (20% fingerprint images). We use the trainset to fine-tune the last layer in deep network models. Then, results are reported on the testset using these resulting models.

Table 1. Description of fingerprint image datasets

ID	Dataset	#Datapoints	#Classes
1	FP-235	3485	235
2	FP-389	6306	389
3	FP-559	10270	559

3.1 Classification results

We obtain the classification accuracy of deep network models in Tables 2, 3, 4 and Figures 6, 7, 8. In tables, the highest accuracy is bold-faced and the second one is in italic.

The VGG16, VGG19, ResNet50, Inception-v3 and Xception give the average classification accuracy of 97.68%, 97.80%, 98.46%, 96.95%, 98.20%, respectively. In the comparison among deep network models, we can see that the ResNet50 and Xception models achieve highest classification accuracy, followed by VGG19 and VGG16.

Maybe **trainbase** dataset is not large enough is the reason why inception-v3 has not achieved as high accuracy as other deep network models. Although the Inception-v3 gives lowest correctness, it is very competitive with the average accuracy of 96.95%.

The classification results are very closed to our previous work in [11], in which we fine-tuned many last layers of deep networks (denoted by FTm-VGG16, FTm-VGG19, FTm-ResNet50, FTm-Inception-v3, FTm-Xception), on the training datasets FP-235, FP-389, FP-559 and then using the resulting models to classify the testset. Nevertheless, the previous approach requires to re-train many last layers in deep network models for a new coming fingerprint image dataset.

We believe that it is possible to train deep network models on such **trainbase** dataset that can be re-used to classify a new coming dataset of fingerprint images without re-training the new deep network model.

Table 2. Overall classification accuracy for FP-235

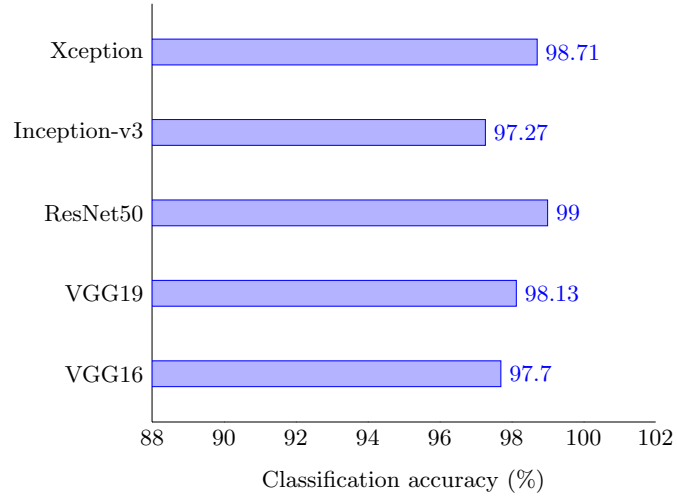
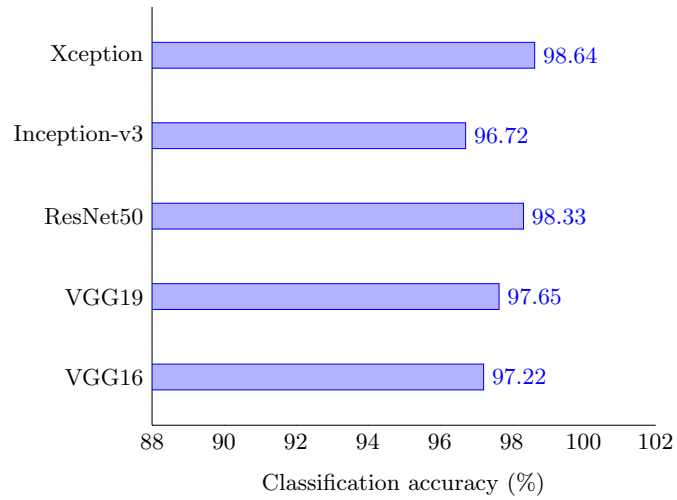
No	Deep network model	Accuracy (%)
1	VGG16	97.70
2	VGG19	98.13
3	ResNet50	99.00
4	Inception-v3	97.27
5	Xception	<i>98.71</i>

Table 3. Overall classification accuracy for FP-389

No	Deep network model	Accuracy (%)
1	VGG16	97.22
2	VGG19	97.65
3	ResNet50	<i>98.33</i>
4	Inception-v3	96.72
5	Xception	98.64

Table 4. Overall classification accuracy for FP-559

No	Deep network model	Accuracy (%)
1	VGG16	98.13
2	VGG19	97.63
3	ResNet50	98.05
4	Inception-v3	96.85
5	Xception	97.26

**Figure 6.** Overall classification accuracy for FP-235**Figure 7.** Overall classification accuracy for FP-389

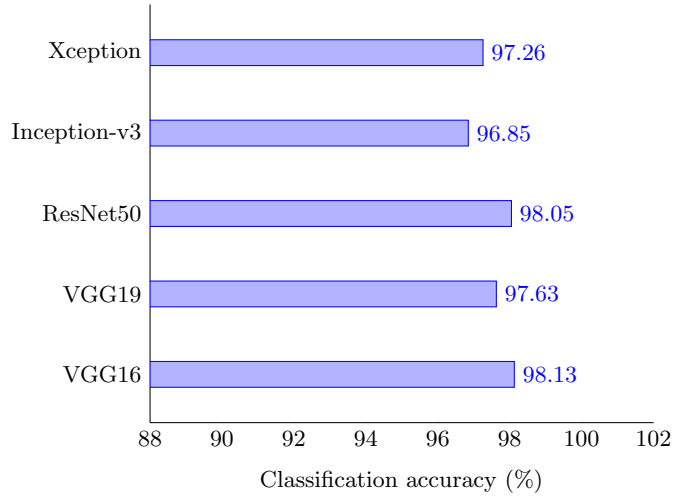


Figure 8. Overall classification accuracy for FP-559

4 Conclusions and future work

We have presented the new proposal to train deep network models for classifying fingerprint images. We collected a real fingerprint image dataset, called **trainbase** from the students at the Can Tho University. After that, we propose to train recent deep networks, including VGG, ResNet50, Inception-v3, Xception, on **trainbase** dataset. For dealing with a new coming fingerprint image dataset, these resulting deep network models can be re-used as the feature extraction and then the train task only performs fine-tuning the last layer in deep network models without re-training a new deep network model from scratch. The empirical test results on three real fingerprint image datasets (FP-235, FP-389, FP-559) show that deep network models achieve at least the accuracy of 96.72% on the testsets. An example of effectiveness is that the ResNet50 models give classification accuracy of 99.00%, 98.33%, 98.05% on FP-235, FP-389 and FP-559, respectively.

In the future, we will enlarge **trainbase** dataset to improve the pre-trained deep network models. We intend to provide more empirical test on large benchmarks.

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